$V \otimes W$
$V \oplus W$
$A \downarrow B$
$\prod$
$\sum_n$

Components

X	object
$f: X \to Y$	arrow
A	category
$F: A \to B \in B^A$	functor i.e. a path
$B^A = \{ A \to B \}$	functor category
$A(X, Y) \in Set$	hom set
$A(-, Y): A \to Set \in Set^A$	hom functor
$\varphi: F \Rightarrow G$	natural transformation
$Nat(F, G) = \{ F \Rightarrow G \}$	

==========

Adjoint

[Stackexchange] How to recognize adjoints: https://math.stackexchange.com/questions/742640/how-to-recognize-adjointness
[Stackexchange] why should I care about adjoint functors: https://math.stackexchange.com/questions/25455/why-should-i-care-about-adjoint-functors
[Mathoverflow] what is an intuitive view of adjoints: https://mathoverflow.net/questions/6551/what-is-an-intuitive-view-of-adjoints-version-1-category-theory

Interpretation 1: weak isomorphism
An adjoint is a **weak isomorphic functor**

	Isomorphic functor G: B → A	**Adjoint** φ**: B → A**
Arrows	$G: \{ -GX \to Y \} \to \{ X \to GY \}$	$\varphi: \{ FX \to Y \} \to \{ X \to GY \}$
	$Gh: X \to GY$	$\varphi h: X \to GY$
Composition		
$f: X' \to X \in A$	$G(h \circ -Gf) = Gh \circ f: X' \to GY$	$\varphi(h \circ Ff) = \varphi h \circ f: X' \to GY$
$g: Y \to Y' \in B$	$G(g \circ h) = Gh \circ Gh: X' \to GZ$	$\varphi(g \circ h) = Gg \circ \varphi h: X \to GY'$

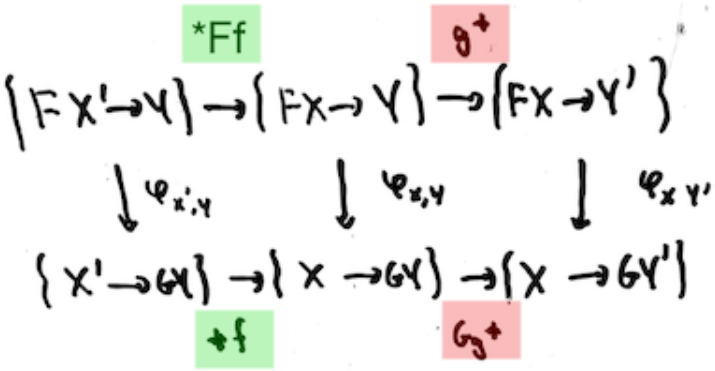
Interpretation 2: universal property natural in both variables
An adjoint is a **universal property natural** in both X and Y

Adjoint
φ: B → A

Universal property
φX,Y: { FX → Y } → { X → GY }

natural in X
φX',Y ∘ *Ff = f* ∘ φX,Y
φX',Y(h ∘ Ff) = φX,Y(h) ∘ f

natural in Y
φX,Y' ∘ g*= Gg* ∘ φX,Y
φX,Y'(g ∘ h) = Gg ∘ φX,Y(h)



Interpretation 3: biuniversal property
An adjoint is a **biuniversal property** i.e. universal in both directions

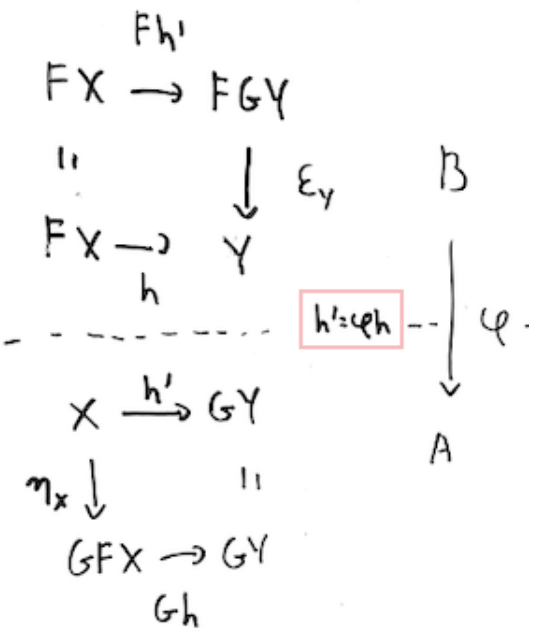
Biunivesal property

	B		A
φ:	{ h: FX → Y }	<=>	{ h': X → GY }

Gh ∘ ηX = h'
h = εY ∘ Fh'

Universal into G
ηX: X → GFX **universal arrow**
φh: X → GY
φh = Gh ∘ ηX

Couniversal into F
ε: FGY → Y co**universal arrow**
-φh': FX → Y
-φh' = εY ∘ Fh'



Definition

Adjoint

Let functors F: A → B, G: B → A

If there exists

$$\varphi_{X,Y}: \{ FX \xrightarrow{B} Y \} \Rightarrow \{ X \xrightarrow{A} GY \}$$

where φ is:

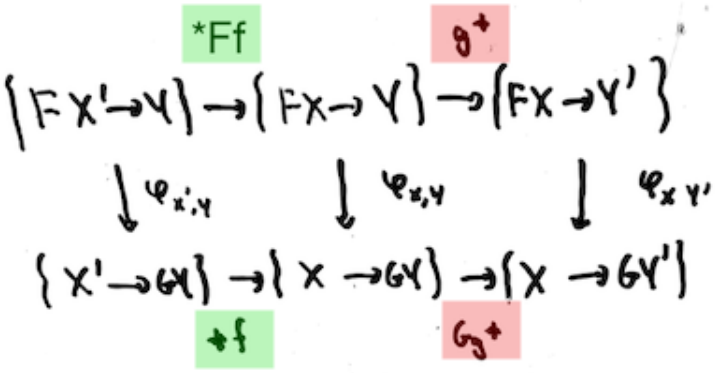
a) **bijective**

b) **natural** in X and Y (as homsets)

$\varphi X',Y \circ {}^*Ff = {}^*f \circ \varphi X,Y$ $\quad f: X \to X'$

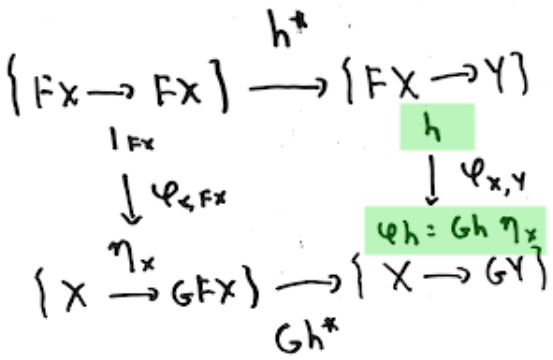
$\varphi X,Y' \circ g^* = Gg^* \circ \varphi X,Y$ $\quad g: Y \to Y'$

Then **<F, G, φ>: A → B** is an **adjunction from A to B.**



Universality φ: {FX→Y} → {X→GY}

$\varphi h = Gh \circ \eta X$



Terminology

B A
φ: { FX → Y } ⇒ { X → GY }

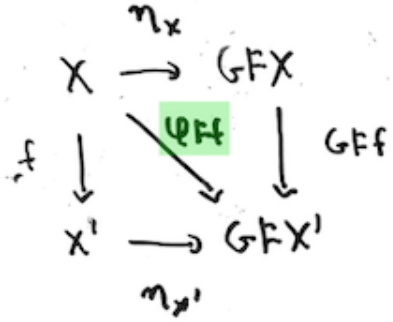
F: A → B	**Left adjoint** of G
G: B → A	**Right adjoint** of F
ηX: X → GFX ∈ A = φ1FX	**unit**
εY: FGY → Y ∈ B = φ^-1(1GY)	**counit**
<F, G, φ>: A → B where φ: { FX → Y } ⇒ { X → GY } B A	**Adjunction** with transformation φ
<F, G, η , ε>: A → B where unit η**: IA ⇒ GF** counit ε**: FG ⇒ IB**	**Adjunction** induced by **units** η and ε

Properties
- **universal/couniversal property**

natural unit aka universal arrow
η: IA ⇒ GF

ηX: X→GFX

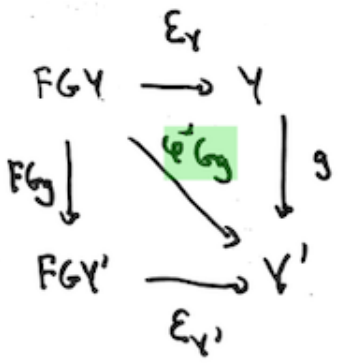
ηX	= φ1FX	definition
ηX ◦ f	= GFf ◦ ηX'	naturality f:X → X'
	= φFf	



natural counit aka couniversal arrow
ε: FG ⇒ IB

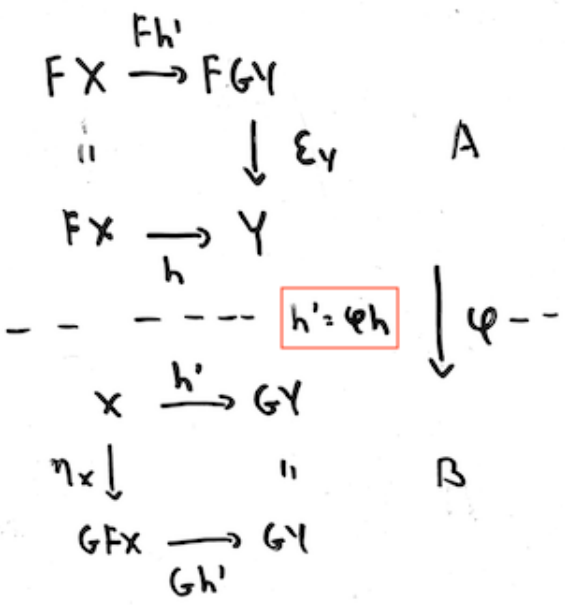
εY: FGY → Y

εY	=	$-\varphi 1GY$	definition
$g \circ \varepsilon Y$	=	$\varepsilon Y' \circ FGg$	naturality g:Y → Y'
	=	$-\varphi Gg$	

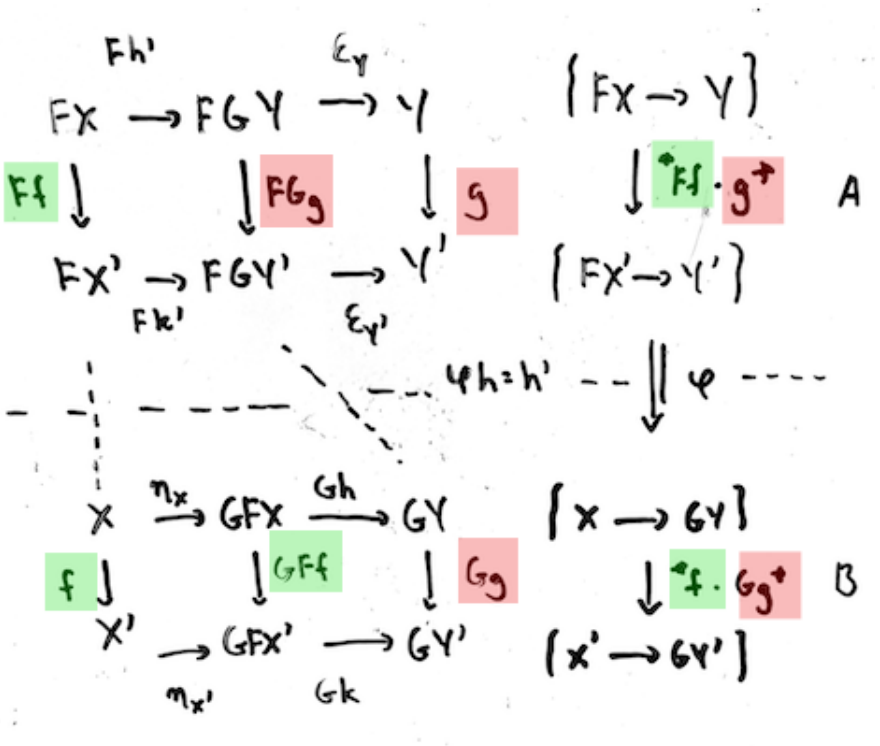


universal properties

φh: X→GY	university property of G
φh = Gh ∘ ηX	h:FX → Y
$-\varphi h'$: FX→Y	university property of F
$-\varphi h'$ = $\varepsilon Y \circ Fh'$	h': X → GY



naturality



Identities
h: FX → Y

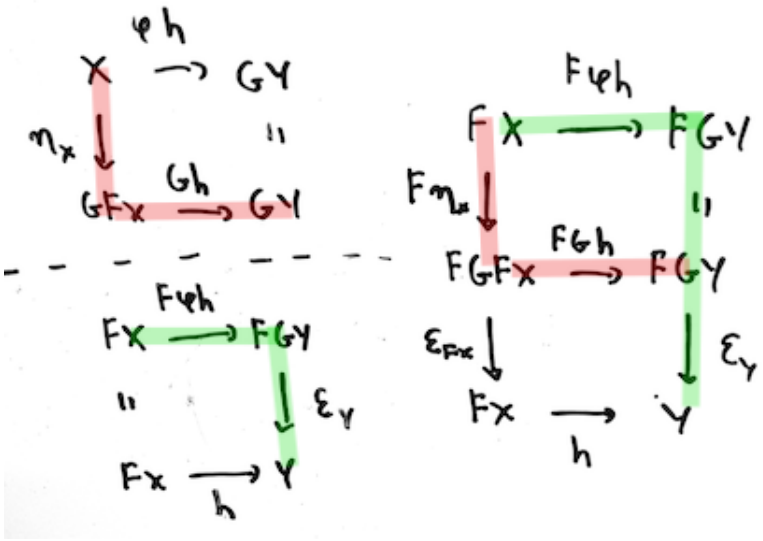
h	=	$-\varphi(\varphi h)$	
	=	$-\varphi(Gh \circ \eta X)$	universality of φ
	=	$\varepsilon Y \circ F(Gh \circ \eta X)$	universality of $-\varphi$
	=	$\varepsilon \mathbf{Y} \circ \mathbf{FGh} \circ \mathbf{F}\eta\mathbf{X}$	
	=	$(h \circ \varepsilon \mathbf{FX}) \circ \mathbf{F}\eta\mathbf{X}$	by naturality of ε
	=	h	
		i.e. preimage of h	
1FX	=	$\varepsilon \mathbf{FX} \circ \mathbf{F}\eta\mathbf{X}$	with h = 1FX

h' = φh

h'	=	$\varphi(-\varphi(h'))$	
	=	$\varphi(\varepsilon Y \circ Fh')$	universality of φ
	=	$G(\varepsilon Y \circ Fh') \circ \eta X$	universality of $-\varphi$
	=	$\mathbf{G}\varepsilon\mathbf{Y} \circ \mathbf{GF}h' \circ \eta\mathbf{X}$	
	=	$G\varepsilon Y \circ (\eta GY \circ h')$	by naturality of η
	=	h'	
1GY	=	$\mathbf{G}\varepsilon\mathbf{Y} \circ \eta\mathbf{GY}$	with h'=1GY

Preimages

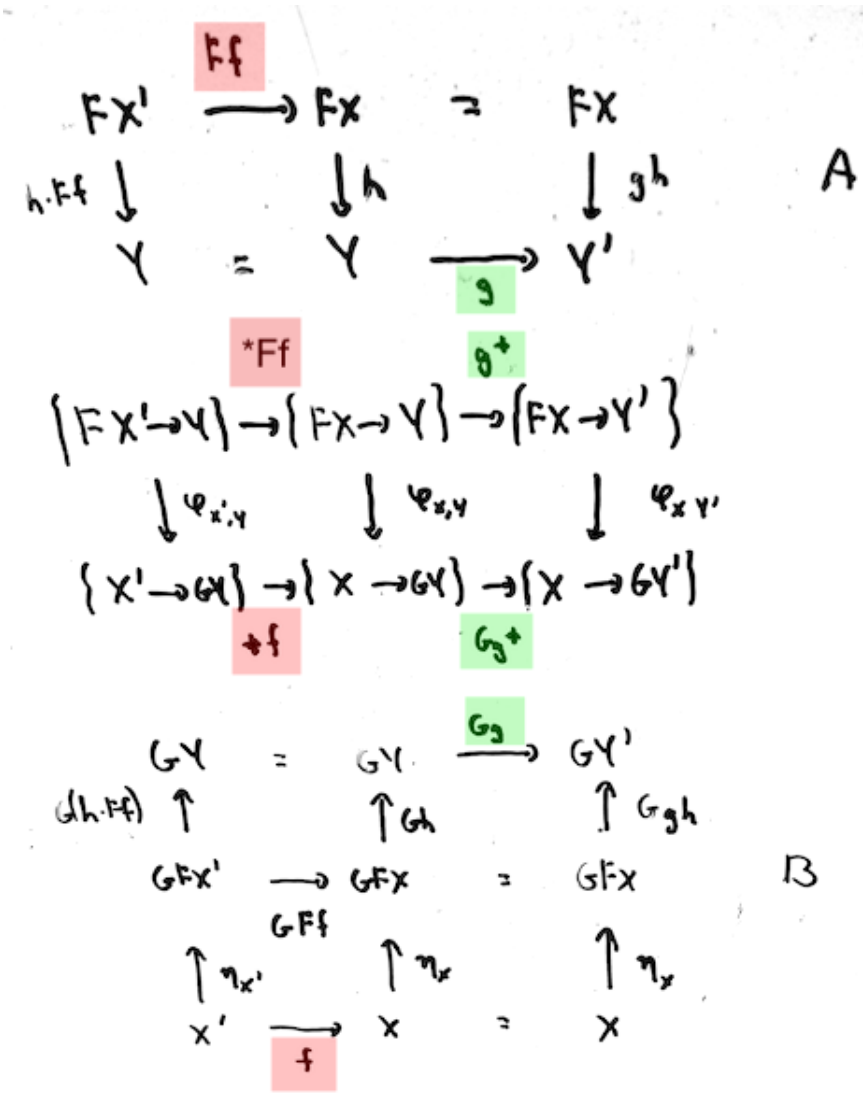
h	=	$\mathbf{-\varphi(Gh \circ \eta X)}$	
	=	$h \circ -\varphi(\eta X)$	by naturality
	=	h	
h'	=	$\varphi(\varepsilon \mathbf{Y} \circ \mathbf{Fh'})$	
	=	$\varphi(\varepsilon Y) \circ h'$	by naturality
	=	h'	



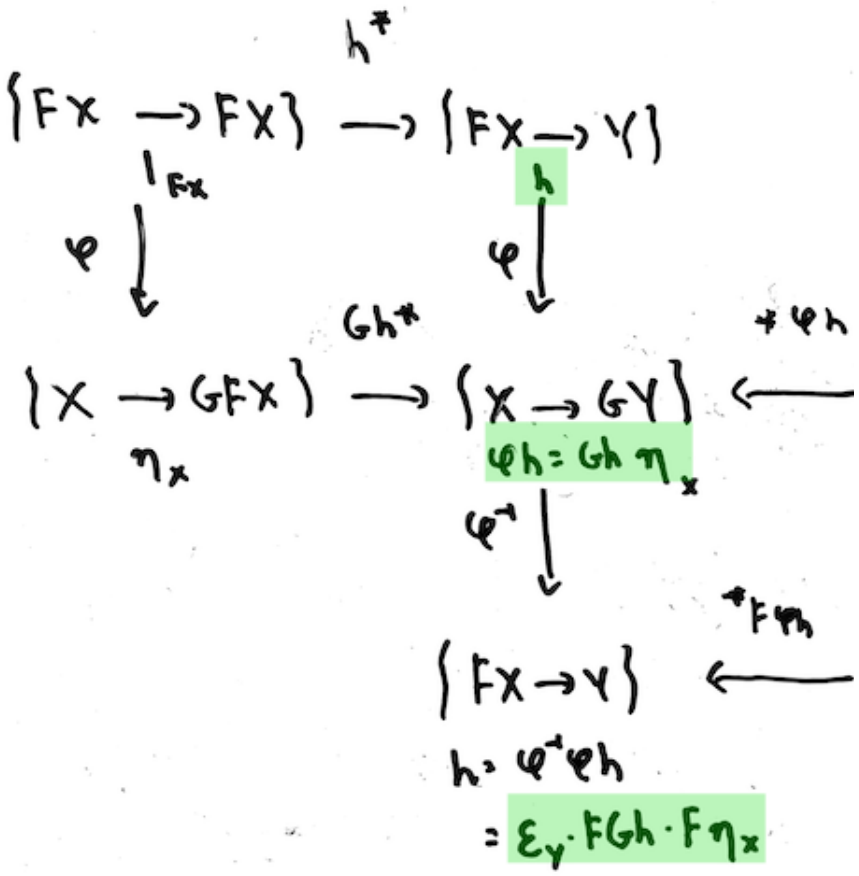
$-\varphi(\eta X)$	=	1GX	by definition
$G\varepsilon \circ \eta G$	=	IG: G ⇒ G	as natural transformation
$\varphi(\varepsilon Y)$	=	1FX	by definition
$\varepsilon F \circ F\eta$	=	IF: F ⇒ F	as natural transformation

- **naturality**

arrow form	in B	$\varphi \circ g^*$	=	$Gg^* \circ \varphi$
	in A	$\varphi \circ {}^*Ff$	=	${}^*f \circ \varphi$
function form	in B	$\varphi(g \circ h)$	=	$Gg \circ \varphi h$
	in A	$\varphi(h \circ Ff)$	=	$\varphi h \circ f$
universal form	in B	$G(g \circ h) \circ \eta X$	=	$Gg \circ (Gh \circ \eta X)$
	in A	$G(h \circ Ff) \circ \eta X$	=	$(Gh \circ \eta X') \circ f$
unit form	in B	$\varepsilon Y \circ g$	=	$FGg \circ \varepsilon Y'$
	in A	$GFf \circ \eta X$	=	$\eta X' \circ f$



Naturality + universality



Corollary: isomorphic functors
If <F, G, φ> and <F', G, φ'> are adjoints, then F and F' are isomorphic

==========
Proof
Universal properties

ηX: X→GFX	universal element of X to GF
η'X: X→GF'X	universal element of X to GF'

=> FX and F'X satisfy the same universal property

=> FX ≅ F'X with isomorphism

φ^-1(η'X): FX→FX' φ'^-1(ηX): FX'→FX

Let

ψ: F ⇒ F'
ψX: FX→FX'
ψX = φ^-1(η'X)

claim: ψ is an isomorphism
i.e. natural + each ψX is an isomorphism

Proof claim:
- isomorphism: by definition

- natural:

for any g: X→Y

ψY ∘ Fg	=	φ^-1(η'Y) ∘ Fg	
	=	φ^-1(η'Y ∘ g)	by composition
	=	φ^-1(GF'g ∘ η'X)	by naturality
	=	F'g ∘ φ^-1(η'X)	
	=	F'g ∘ ψX	

QED
==========

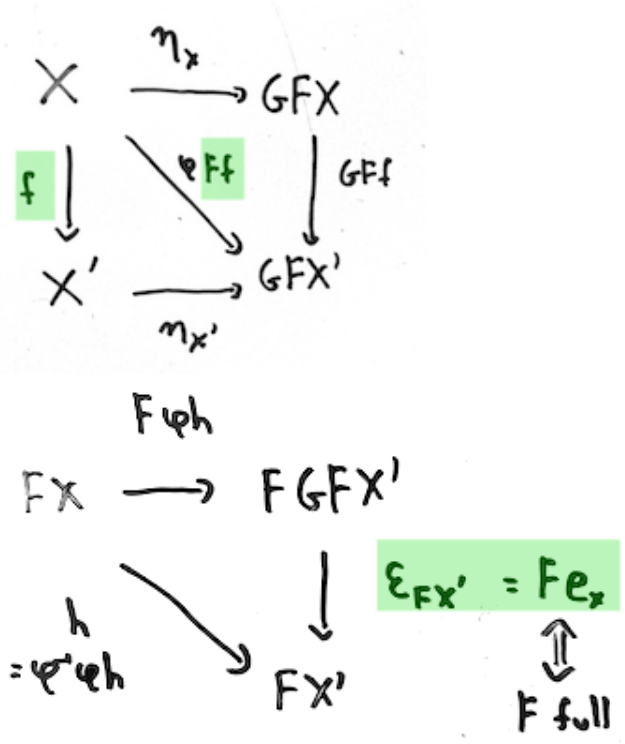
Proposition: preservation of isomorphisms
Let <F, G, φ>

φX.Y **preserves isomorphisms** <=> **unit ηX: X → GFX isomorphic**
i.e. h: FX → Y isomorphism
=> φh isomorphism

-φX.Y **preserves isomorphisms** <=> **counit** εY: FGY → Y **isomorphic**

Proof: universal property preserves isomorphisms

Full/faithful
Adjoints are a form of isomorphism between categories.
Hence it provides a way to define **full** and **faithful** in terms of **natural transformations**.



Proposition: full/faithful

F is **faithful** <=> η is **monic**

F is **full** <=> ε**FX = FeX** for any X
<=> η is **split epic**
ηX ◦ eX = 1GFX

G is **faithful** <=> ε is **epic**

G is **full** <=> η**GY = GnY** for any Y
<=> ε is **split monic**
εY ◦ nY = 1FGY

Hence

F is **full** and **faithful** <=> η is a natural **isomorphism**

G is **full** and **faithful** <=> ε is a natural **isomorphism**

==========
Proof:

- **faithful +** η

ηX ◦ f = ηX' ◦ GFf by naturality
= φFf

Hence

η is monic <=> φFf = φFf' implies f =f '
<=> Ff = Ff' implies f = f' by 1-1 of adjoint
i.e. faithful

- **full** + η

Let h: FX → FX'. Use the adjoint to **map** it into F:

h = -$\varphi\varphi$h
= εFX' ◦ Fφh
= F(eFX' ◦ φh) if εFX' = FeX for some eX': GFX' → X'

=>

F is **full** <=> εFX = FeX for some eX: GFX → X
<=> 1GFX = η**X** ◦ **eX** by **naturality** ηX ◦ eX = GεFX ◦ ηGFX = 1GFX
i.e. ηX is split epic

- isomorphism

By general property, **monic + split epic** implies isomorphic
i.e.

ηX = (ηX ◦ eX) ◦ ηX
= ηX ◦ (eX ◦ ηX)
ηX = ηX ◦ 1
=> (eX ◦ ηX) = 1 since ηX is monic

QED
==========

e.g. **forgetful functor U**: faithful

Vect Set

{ Free X → V } <=> { X → UV }

U: Vect → Set	<=>	εV: Free(UV) → V
forgetful is faithful		**interpretation** is surjective
Free: Set → Vect	<=>	ηX: X → UFree(X)
free construction is faithful		**free injection** is injective

e.g. **injection functor I**: full + faithful

Ab **Grp**
{ G/[G, G] → A } <=> { G → IA } adjoint

I: Ab → Grp	<=>	εA: A/ [A, A] → A
injection is full + faithful		εA(a) = a
		injective homomorphism is isomorphic
Abelian: Grp → Ab	<=>	ηX: G → I(G/[G, G])
Abelian(G) = G/[G, G]		ηX(g) = g/[G, G]
Abelianization is full		Abelianization map is surjective

e.g. **Δ**X: A → AxA

A x A A
{ **Δ**X → (Y, Y') } <=> { X → YxY' }

Δ is faithful	<=>	ηX: X → XxX
		monic

==========

--- Binaturality ---

Binaturality
Adjoint is a **universal property** that is **binatural**
i.e. **natural in both** categories A and B

Theorem: naturality
Let bijection φX,Y: {FX→Y} → {X→GY} with functors F: A → B, G: B → A

φ is a **universal property** of FX	<=>	φ is **natural** in Y (as homsets)
φh = Gh ∘ φ1	φ(h∘1) ∈ φX,Y ∘ h*	φX,Y' ∘ g* = Gg* ∘ φX,Y
	Gh ∘ φ1 ∈ Gh* ∘ φX,FX	

φ is an **adjoint**	<=>	φ is **natural** in X and Y (as homsets)
	definition	φX,Y' ∘ g* = Gg* ∘ φX,Y
		φX,Y∘ *Ff = *f ∘

$\varphi X',Y$

<=> φ is a **universal property natural** in X

$\varphi X\ h = Gh \circ \eta X$ universal

$\varphi X'\ h \circ f = \varphi X\ (h \circ Ff)$ natural in X

<=> **universal arrow** is natural

$(Gh \circ \varphi 1FX) \circ f = G(h \circ Ff) \circ \varphi 1FX'$ $\eta X \circ f = GFf \circ \eta X'$

==========

Proof

- **case**: natural in X and Y <=> universal natural in X)

natural in Y
<=> universal property

natural in X
<=> $*f \circ \varphi X,Y = \varphi X',Y \circ *Ff$
<=> $\varphi X,Yh \circ f = \varphi X',Y(h \circ Ff)$
i.e. natural in X

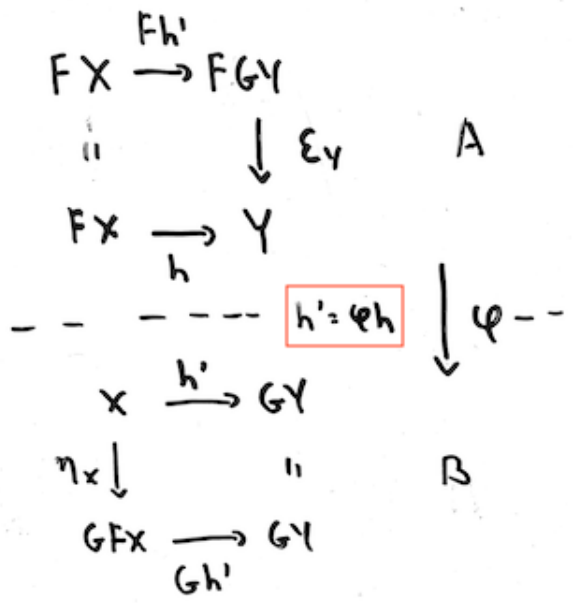
QED

==========

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--- Biuniversality ---

Biuniversality
Adjoint is **binuniversal**
i.e. universal in both directions φ and $-\varphi$.



Theorem: biuniversal
Let bijection φ: {FX→Y} → {X→GY} with functors F: A → B, G: B → A

φ is adjoint	<=>	φ is **biuniversal**	
		$\varphi g = Gg \circ \eta X$	universal
		$-\varphi f = \varepsilon Y \circ Ff$	couniversal

==========
Proof

natural in Y	<=>	universal
natural in X	<=>	couniversal

QED
==========

A universal property is determined completely by the **universal arrows** i.e. **units (Yoneda)**.

Theorem: biuniversal by units
Let bijection φ: {FX→Y} → {X→GY} with functors F: A → B, G: B → A

φ is **biuniversal** i.e. **adjoint**		<=>	**natural**
units η: IA ⇒ GF, ε: FG ⇒ IB			
$\varphi g = Gg \circ \eta X$	universal	$\varphi(\varepsilon Y) = G\varepsilon Y \circ \eta GY$	**IG = Gε ∘ ηG**
$-\varphi f = \varepsilon Y \circ Ff$	couniversal	$-\varphi(\eta X) = \varepsilon FX \circ F\eta X$	**IF = εF ∘ Fη**

==========
Proof
- adjoint => natural units

$$
\begin{aligned}
1GY &= \varphi(-\varphi(1GY)) \\
&= G(-\varphi(1GY)) \circ \eta GY \\
&= G\varepsilon Y \cdot \eta GY
\end{aligned}
$$

=> IG = Gε ∘ ηG

and

φ is adjoint
=> universal arrow ηX is natural

Likewise with 1FX.

- natural units => adjoint
Idea: just combine with unit + use naturality to invert
For any h: X → GY

$\varphi(\varepsilon Y \circ Fh) = G(\varepsilon Y \circ Fh) \circ \eta X$ **NOTE**: $\varepsilon Y \circ Fh = -\varphi h$
$= G\varepsilon Y \circ (GFh \circ \eta X)$
$= G\varepsilon Y \circ \eta GY \circ h$ natural
$= h$
=> φ is **surjective**

Conversely, for any $\varphi g = Gg \circ \eta X$

$\varepsilon Y \circ F\varphi g = \varepsilon Y \circ FGg \circ F\eta X$ NOTE: $\varepsilon Y \circ F\varphi g = -\varphi(\varphi g)$
$= g \circ \varepsilon FX \circ F\eta X$ natural
$= g$
=> φ is **injective**

=> φ is **bijective**, hence a universal property.

Likewise for $-\varphi$

QED
==========

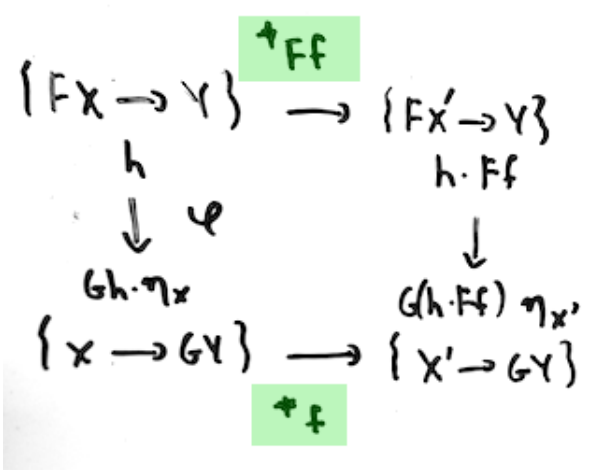
==========
======
--- Characterization ---

Extension
A universal properties can be **extended** to be **binatural** i.e. natural in both directions

Proposition: binatural extension
For each X, let bijection $\varphi X,Y$: {FX→Y} → {X→GY} with functor G: B → A

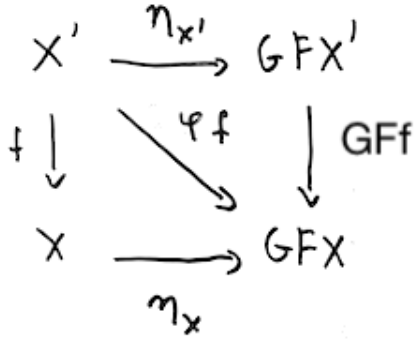
Then there is exactly one way to extend F to a **functor F: A → B** such that φ is also **natural in A.**
For any f: X' → X

φ is **natural in A** <=> **Ff: FX' → FX**
$\varphi h \circ f = \varphi(h \circ Ff)$ **Ff = -φ(**η**X** ∘ **f)**
(Gh ∘ ηX) ∘ f = G(h ∘ Ff) ∘ ηX'



In such a case, Ff is precisely the natural arrow for the **universal arrows** η**X**:

η**X** ◦ **f** = **GFf** ◦ η**X'**



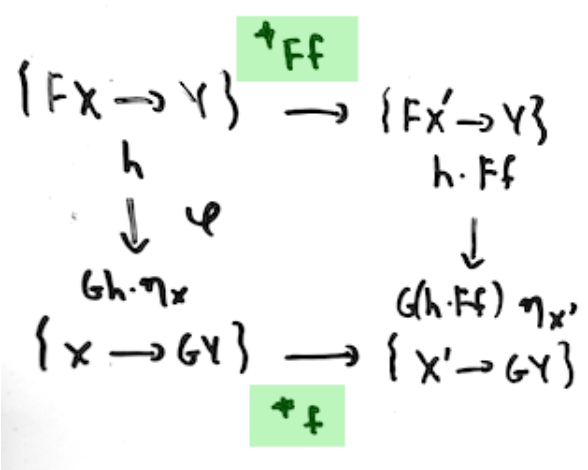
Corollary: extension
A universal property can be uniquely extended to be **binatural**
That is, an **adjoint**.

==========
Proof:

<=) Pick any h:FX → Y

φh ◦ f	=	(Gh ◦ ηX) ◦ f	
	=	Gh ◦ φFf	by assumption Ff = φ^-1(ηX ◦ F)
	=	φ(h ◦ Ff)	by naturality in B

=> φ is natural in A



=>) Suppose φ is natural in A.
Then

$\varphi(Ff \circ 1FX) = \varphi 1FX \circ f$
$= \eta X \circ f$

=> $Ff = \varphi^{-1}(\eta X \circ 1)$

- naturality of ηX

$\eta X \circ f = \varphi 1FX \circ f$
$= \varphi(1FX \circ Ff)$
$= GFf \circ \eta X'$

QED
==========

Characterization
Theorem: characterization

The following are equivalent

1) <F, G, φ>: A → B adjunction

2) universal property + **natural**:

F: A → B, G: B → A — functor
$\varphi X,Y$: {FX → Y} → {X → GY} — universal properties natural in both X and Y

3) universal property + **extension**:

G: B → A — functor
$\varphi X,Y$: {FX → Y} → {X → GY} — universal properties defined for each X

Adjoint if F is extended to a functor

functor F: A → B
Ff: FX→FX'
$Ff = -\varphi(\eta X \circ f)$ i.e. unique value such that
$f = GFf \circ \eta X$

4) universal property (via **universal element**) + **natural**:

F: A → B, G: B → A	functors
η: IA ⇒ GF	
ηX: X→GFX	universal element of {X→G_}

5) universal property (via **universal element**) + **extension**:

G: B → A	functor
FX	object for each X
ηX: X→GFX	universal element of {X→G_}

Adjoint if F is extended to a functor

functor F: A → B	
Ff: FX→FX'	
Ff = f'	unique f' such that f = Gf' ηX

6) **natural unit + counit**:
Natural units determine a universal property

natural units
η: IA ⇒ GF
ε : FG ⇒ IB

Identity
εF ∘ Fη = IF
Gε ∘ ηG = IG

In all cases, the adjunction is defined from the unit η: IA ⇒ GF by

adjunction
φ: {FX→Y} → {X→GY}
φf = Gf ∘ ηX

Corollary (Counit)
Adjoints can be defined in terms of counits
i.e.
The following are equivalent

1) <F, G, φ> is an adjunction

2) via **counit**:

F, G	functors
ε: FG ⇒ IB	natural
εY: FGY → Y	couniversal element of {F_→Y}

3) via **unit**:

F	functor
G: B → A	object function
εY: FGY → Y	natural
g ∘ εY = εY' ∘ FGg	i.e. natural in Y ∈ obj(B)

This induces a functor G

functor G
G: B → A
Gg: GY'→GY
Gg = g' unique g' such that g = εY ◦ Fg'

In all cases, the inverse adjunction is defined from the counit εY: FGY → Y
by

adjunction
φ^-1: {X→GY} → {FX→Y}
φ^-1 = εY ◦ Fg

Limit
Adjuncts preserve **limits** by **1-1 correspondence** of arrows.

Theorem: limit preservation
Adjusts preserve limits
i.e. Let <F, G, φ>: A → B be an adjunction
For any shape T: J → A, S: J → B

Colim FT = F (Colim T) if Colim T exists
universal **Fu: FT => ΔF(Colim T)** universal u: T ⇒ ΔColim T

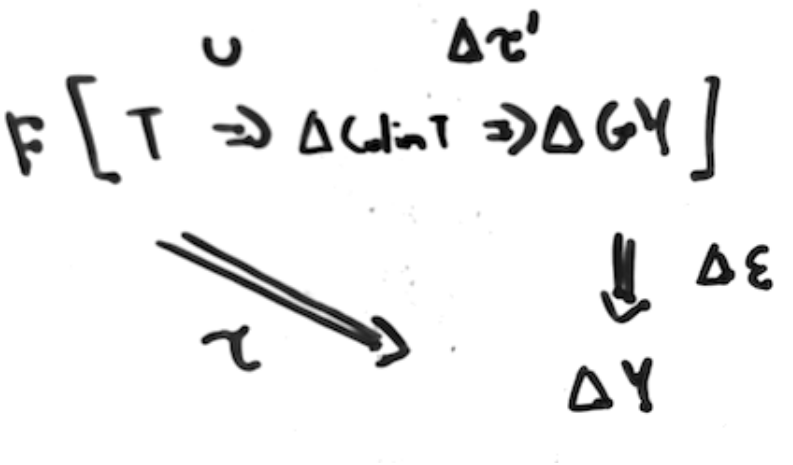
Lim GS = G (Lim S) if Lim S exists
universal **Gv: Δ G (Lim S) ⇒ GS** universal v: ΔLim S ⇒ S

==========
Proof
Idea: adjoint preserves **cones**

Let τ: FT ⇒ ΔY be any cone

$\varphi\tau$: T ⇒ ΔGY
$\varphi\tau$j = φ(τj)
=> unique τ': Colim T → GY
$\varphi\tau$ = Δτ' ◦ u by colim

=>
τ = -φ ($\varphi\tau$)
= ΔεY ◦ F$\varphi\tau$
= ΔεY ◦ FΔτ' ◦ Fu
= **Δ(εY ◦ Fτ')** ◦ **Fu** since FΔX = ΔFX for any X



- **uniqueness**: $\varepsilon Y \circ F\tau'$: F Colim T → Y is unique
If there is another ρ: F Colim T → Y such that

$\tau = \Delta\rho \circ Fu$

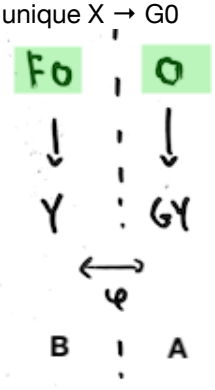
$$
\begin{aligned}
\Rightarrow \tau &= \Delta\rho \circ Fu \\
&= \Delta(\varepsilon Y \circ F\varphi\rho) \circ Fu \qquad \text{by adjoint } \rho = -\varphi(\varphi\rho) = \varepsilon Y \circ F\varphi\rho \\
&= \Delta\varepsilon Y \circ F(\varphi\rho \circ u)
\end{aligned}
$$

QED
==========

Corollary

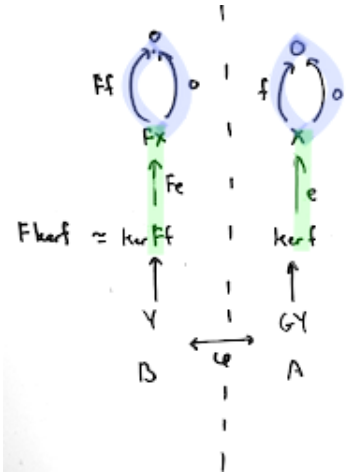
initial/terminal
F0 is **initial** — if 0 is initial
unique F0 → Y fo

G0 is **terminal** — if 0 is terminal
unique X → G0



monic/epic
Ff: FX → FX' is **epic** — if f: X → X' is epic
Gg: GY → GY' is **monic** — if g: Y → Y' is monic

kernel
ker Ff ≅ F(ker f)
Coker Gg ≅ G(Coker g)



product/coproduct

FX x FX' ≅ F (X x X') if X x X' ∈ A exists

projections

Fp: FX x FX → FX

Fq: FX x FX' → FX'

projections (p,q): X x X' → (X, X')

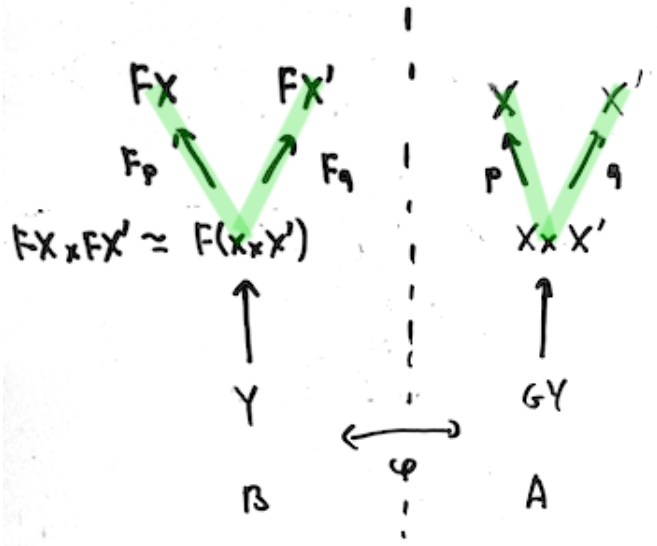
GY ⊔ GY' ≅ G (Y ⊔ Y') if Y ⊔ Y' ∈ B exists

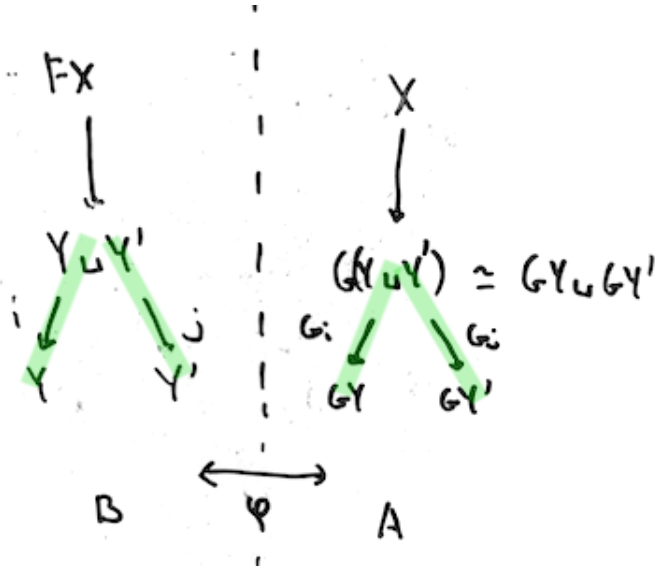
injections

Gi: GY → GY ⊔ GY'

Gj: GY → GY ⊔ GY'

injections (i, j): (Y, Y') → Y ⊔ Y'





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--- Equivalence ---

Equivalence is where F: A → B is an **isomorphism** with a **subcategory FA** ⊆ B.

Theorem: characterization
Let F: A → B

Equivalent statements:
1) F is an **equivalence** of categories

2) <F, G, ε, η> are **equivalent adjoints**

3) F is **full** and **faithful**
Each Y ∈ B is **isomorphic** to at least one object of FA.
i.e. can define GY ∈ FA such that **Y ≅ FGY**

Corollary: reflection
Let A' ⊆ A be a **full subcategory** of A.
If each X ∈ A is isomorphic to some FX ∈ A' with FX' = X' then the **reflection F** is an **equivalence**.
i.e.

<F, K, 1: IA' → IA', η: IA → KF>

==========
Proof:

2 => 3)
- full and faithful:
unit ηX: X → GFX is an isomorphism
=> F is faithful and full

- Y ∈ B
εY: FGY → Y is an isomorphism
=> Y ≅ FGY ∈ FA

3 => 2)
Idea: just construct a 1-1 correspondence with subcategory FA

Let Y ∈ B
Define GY ∈ FA by

εY: FGY → Y isomorphism

claim: εY is a unit of { X → GY } → { FX → Y }

proof claim:
Let h: FX → Y

=> εY^-1 ◦ h: FX → FGY

=> Fh' = εY^-1 ◦ h for some h: X → GY since F is full
h' is unique since F is faithful

=> 1-1 relationship h: FX → Y <=> h': X → GY
QED

Then

<F, G> an adjoint	by extending G to a functor
εY: FGY → Y an **isomorphism**	by definition
ηX: X → GFX an **isomorphism**	inverse defined by
	εFX ◦ FηX = 1FX
	=> ηX = F^-1(εFX^-1)

F^-1 exists since F is faithful

ηX^-1 = F^-1(εFX)

QED
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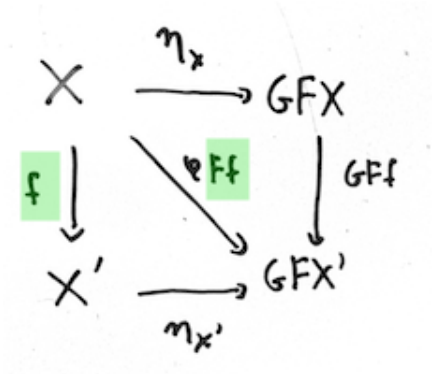
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----- OLD -----

- **faithful**:
Let Ff: FX → FX'. **Invert** it to f with the adjoint:

φFf	=	GFf ◦ ηX	
	=	ηX' ◦ f	by naturality

=>
F is **faithful** <=> ηX is **monic**



- **full**:
Let h: FX → FX'. Use the adjoint to **map** it into F:

h = -$\varphi\varphi$h
= εFX' ◦ Fφh
= F(eFX' ◦ φh) if εFX' = FeX for some eX': GFX' → X'

=>

F is **full** <=> εFX = FeX for some eX: GFX → X
<=> 1GFX = η**X** ◦ **eX** by **naturality** ηX ◦ eX = GεFX ◦ ηGFX = 1GFX
i.e. ηX is split epic

Hence

Adjoint
Idea
A canonical notion of an **isomorphic functor F: A → B**
i.e. a **1-1 correspondence of morphisms** between A and B.

If F: A → B is bijective, then there is a 1-1 correspondence:

B		**A**
Fh: FX → FX'	<=>	h: X → X'

or for symmetry with G = F^-1

B		**A**
Fh: FX → Y	<=>	h: X → GY
Fh = εY ◦ Fh		h = G(Fh) ◦ ηX
unit 1Y = εY: FGY → Y		**unit** 1X = ηX: X → GFX

An **adjoint** is a generalization of this to any G: B → A and offsets with a 1-1 correspondence:

units
η: IA ⇒ GF ε: FG ⇒ IA

B		**A**
h': FX → Y	<=>	h: X → GY
h' = εY ◦ Fh		h = Gh' ◦ ηX

-

	Functor G: B → A	**Adjoint φ: B → A**
Arrows	G: { Y → Y' } → { GY → GY' }	φ: { FX → Y } → { X → GY }
	Gh: GY → GY'	φh: X → GY
Composition	G(h ∘ f) = Gh ∘ Gf: GX → GY	φ(h ∘ Ff) = φh ∘ f: X' → GY
	G(g ∘ h)= Gh ∘ Gh: GX → GZ	φ(g ∘ h) = Gg ∘ φh: X → GY'
	where f: X → Y g: Y' → Z	where f: X' → X g: Y → Y'